CHAPTER 05: MANAGING THE TEST ACTIVITIES

1. Core Syllabus Concepts & In-Depth Notes

5.1 Test Planning FL-5.1.1 FL-5.1.2 FL-5.1.3

Formal Definition: A **test plan** describes the test objectives, resources, and processes for a test project. It guides testers' thinking, forces confrontation with future challenges (risks, schedules, costs), and serves as a vehicle of communication with stakeholders.

5.1.1 Purpose and Typical Content of a Test Plan (ISO/IEC/IEEE 29119-3)

Core Purposes of Test Planning:

- Documents means and schedule for achieving test objectives.
- Ensures performed activities meet established criteria.
- Serves as a means of communication with team and stakeholders.
- Demonstrates adherence to test policy and strategy (or explains deviations).

Typical Sections in a Test Plan:

- **Context:** Test scope, objectives, test basis.
- **Assumptions & Constraints** of the project.
- **Stakeholders:** Roles, training and hiring needs.
- **Risk Register:** Product and project risks.
- **Test Approach:** Levels, types, techniques, criteria, tools.
- **Budget and Schedule:** Milestones, resource allocation.

5.1.2 Tester's Contribution to Iteration and Release Planning FL-5.1.2 (K1)

Release Planning (Long-Term Horizon):
Looks ahead to product release; refines the product backlog and splits large user stories into smaller ones. Testers contribute by:
• Writing testable user stories and acceptance criteria.
• Participating in project and quality risk analyses.
• Estimating testing effort associated with user stories.
• Defining the overall test approach across all iterations.

Iteration Planning (Sprint Horizon):
Looks ahead to a single iteration (sprint backlog). Testers contribute by:
• Detailed risk analysis of individual user stories.
• Assessing testability of user stories.
• Breaking down user stories into testing tasks.
• Estimating test effort for all testing tasks.
• Identifying functional and non-functional aspects.

5.1.3 Entry Criteria and Exit Criteria FL-5.1.3 (K2)

Dimension	Entry Criteria (Definition of Ready - DoR)	Exit Criteria (Definition of Done - DoD)
Formal Purpose	Defines **preconditions** for undertaking a given test activity. If not met, activities are harder, costlier, and riskier.	Defines what must be achieved to declare an activity, test level, or release **completed**.
Typical Criteria Examples	• Availability of testware (testable requirements, test cases) • Availability of test environment, test data, tools, staff • Initial quality gate: all smoke tests have passed	• Thoroughness measures: achieved coverage, defect density • Residual defect thresholds: zero unresolved critical bugs • Binary criteria: all regression tests automated, execution 100%
Budget/Time Realities	Delaying entry is preferable to testing against broken builds.	Running out of time/budget is a valid exit criterion *if stakeholders review and accept the residual risk*.

5.1.4 Test Effort Estimation Techniques FL-5.1.4 (K3)

Test effort estimation predicts the amount of work required to achieve test objectives. Estimating smaller decomposed tasks is significantly more accurate than estimating large tasks. The syllabus defines four core techniques:

1. Estimation Based on Ratios (Metrics-Based):

Collects figures from previous organizational projects to establish standard ratios.

Worked Example: If historical development-to-test effort ratio is **3 : 2**, and current project dev effort is expected to be **600 person-days**:

Test Effort = 600 × (2 / 3) = 400 person-days

2. Extrapolation (Metrics-Based):

Measurements gathered early in the current project are extrapolated forward using a mathematical model. Highly suitable for iterative/Agile SDLCs.

Worked Example: If actual test effort in sprints 1, 2, and 3 was 18, 22, and 20 days, sprint 4 is estimated as the average:

Sprint 4 Effort = (18 + 22 + 20) / 3 = 20 person-days

3. Wideband Delphi (Expert-Based):

An iterative consensus technique where experts make estimates in isolation. Results are collated. If deviations exceed agreed boundaries, experts discuss and submit revised estimates anonymously until consensus is reached.

- **Planning Poker:** An Agile variant of Wideband Delphi using cards representing story points or effort size.

4. Three-Point Estimation (PERT / Expert-Based):

Experts provide three estimates:

- *a* = Most optimistic estimate
- *m* = Most likely estimate
- *b* = Most pessimistic estimate

Weighted Mean Estimate (*E*) and Standard Deviation (*SD*):

$$E = (a + 4m + b) / 6 \quad | \quad SD = (b - a) / 6$$

Worked Permutation Example: Three-Point Estimation Calculation

Scenario: Three experts estimate testing a payment integration. Optimistic *a* = 6 hours, Most likely *m* = 9 hours, Pessimistic *b* = 18 hours.

Step 1 (Calculate Weighted Estimate E):

- *E* = (6 + 4(9) + 18) / 6 = (6 + 36 + 18) / 6 = 60 / 6 = **10 person-hours**

Step 2 (Calculate Measurement Error / Standard Deviation SD):

- *SD* = (18 − 6) / 6 = 12 / 6 = **2 person-hours**

Final Reported Estimate: **10 ± 2 person-hours** (High confidence interval: between **8** and **12** hours).

5.1.5 Test Case Prioritization FL-5.1.5 (K3)

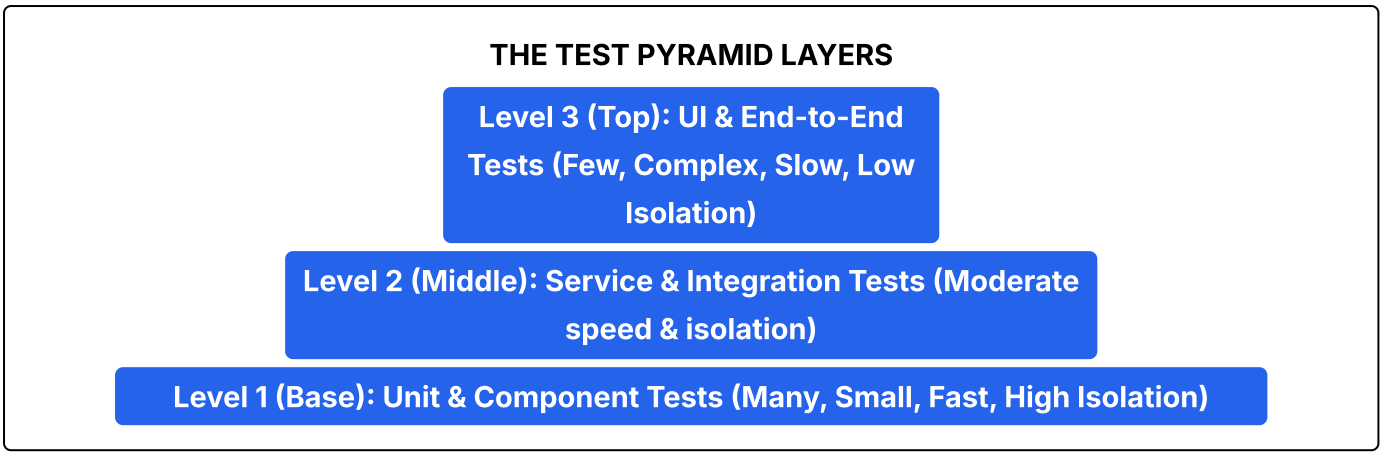
Once test cases and procedures are assembled into test suites, they are arranged into an optimal **test execution schedule** based on one of three common prioritization strategies:

Prioritization Strategy	Ordering Mechanism	Exam Scenario Application
Risk-Based Prioritization	Order of execution is driven by product risk analysis. Test cases covering the highest risk levels are executed first.	Critical security and core payment transactions executed on day 1 of test run.
Coverage-Based Prioritization	Order is based on coverage (e.g. statement coverage). In *Additional Coverage Prioritization*, the test achieving highest coverage runs first, followed by tests giving highest marginal additional coverage.	Maximizing code or requirements coverage rapidly when time windows are limited.
Requirements-Based Prioritization	Order is driven by stakeholder business priority of the requirements traced back to the test cases.	Client prioritizes "Must-Have" features per MoSCoW criteria.

CRITICAL EXAM OVERRIDE: Test Case Dependencies & Resource Constraints:
Even if a test case has the highest risk or requirement priority, **if it has technical dependencies on a lower-priority test case, the lower-priority test case MUST be executed first!** Furthermore, execution order must account for resource availability (specialized hardware, test environments, or SMEs available only in specific time windows).

5.1.6 The Test Pyramid FL-5.1.6 (K1)

Core Concept (Cohn 2009): The test pyramid models test granularity, test isolation, and effort allocation. It illustrates that different test objectives are supported by different levels of automation.



Key Pyramid Mechanics (Syllabus Errata Precision):

- **Bottom Layer (Unit):** Tests are small, highly isolated, fast, and test narrow functionality. Many tests are required to achieve reasonable coverage.
- **Top Layer (UI / E2E):** Tests are complex, high-level, slow, and check large end-to-end functionality. Only a few are needed for coverage.
- **Exam Rule:** *The higher the layer, the lower the test granularity, the lower the test isolation, and the higher the execution time!*

5.1.7 The Testing Quadrants FL-5.1.7 (K2)

Origin & Purpose (Brian Marick 2003, Crispin 2008): The testing quadrants model groups test levels with appropriate test types, activities, and techniques in Agile software development. It enables test management to ensure that all test types are represented and communicate testing scope across business and technical team members.

The model combines two orthogonal dimensions: (1) **Business Facing vs. Technology Facing**, and (2) **Supporting the Team (guide development) vs. Critiquing the Product (evaluating against expectations)**:

Quadrant Q1: Technology Facing • Support Team

• **Test Types:** Component testing (unit) and component integration testing.
• **Characteristics:** Highly automated; embedded directly into the Continuous Integration (CI) build pipeline.
• **Goal:** Guides developer coding and verifies fundamental code correctness.

Quadrant Q2: Business Facing • Support Team

• **Test Types:** Functional tests, examples, user story tests, user experience prototypes, API testing, simulations.
• **Characteristics:** Validates user story acceptance criteria; can be manual or automated.
• **Goal:** Guides business-aligned feature development.

Quadrant Q3: Business Facing • Critique Product

- **Test Types:** Exploratory testing, usability testing, User Acceptance Testing (UAT).
- **Characteristics:** User-oriented, human evaluation; predominantly manual.
- **Goal:** Evaluates actual user experience and business fitness in realistic scenarios.

Quadrant Q4: Technology Facing • Critique Product

- **Test Types:** Smoke tests and non-functional tests (performance efficiency, reliability, security, maintainability).
- **Exception:** Usability is in Q3; all other non-functional testing resides in Q4!
- **Characteristics:** Tool-driven; heavily automated.

Testing Quadrants Master Summary Table:

Quadrant	Orientation & Purpose	Included Test Types & Activities	Execution Mode
Q1	Technology-facing • Support team	Component tests, unit integration tests	Automated (CI)
Q2	Business-facing • Support team	Functional tests, user story tests, API tests, prototypes	Manual or Automated
Q3	Business-facing • Critique product	Exploratory testing, usability testing, UAT	Predominantly Manual
Q4	Technology-facing • Critique product	Smoke tests, performance, load, security, reliability	Predominantly Automated

1.2 Risk Management

Formal Definition (ISO 31000): A **risk** is a potential event, hazard, threat, or situation whose occurrence causes an adverse effect. The test approach where activities are selected, prioritized, and managed based on risk analysis and control is called **risk-based testing**.

5.2.1 Risk Definition and Attributes `FL-5.2.1 (K1)`

A risk is defined by two fundamental factors:

- **Risk Likelihood:** The probability of occurrence ($0 < \text{Likelihood} < 1$).
- **Risk Impact (Harm):** The potential damage or consequence if the event occurs.
- **Risk Level:** The quantitative or qualitative measure of risk: `Risk Level = Likelihood × Impact`

5.2.2 Project Risks vs. Product Risks FL-5.2.2 (K2)

Dimension	Project Risks (Management & Delivery)	Product Risks (Quality Characteristics)
Definition	Risks related to the management and control of the project that threaten schedule, budget, or scope.	Risks related to product quality characteristics (ISO 25010) that threaten customer satisfaction.
Organizational Issues	Delays in work product delivery, inaccurate estimation, budget cuts.	Poor architecture, inefficient algorithms, high maintenance costs.
People Issues	Staff shortages, skill deficits, team conflict, communication failure.	User dissatisfaction, poor user experience (UX).
Technical Issues	Scope creep, inadequate tool support, unstable environments.	Missing functionality, runtime crashes, calculation errors, security vulnerabilities.
Supplier Issues	Third-party vendor delivery failure, contractor bankruptcy.	Data corruption, third-party API integration timeouts.
Consequences	Missed deadlines, project budget overrun, reduced release scope.	Financial loss, reputation damage, help desk overload, legal penalties, physical injury/death.

EXAM CLASSIFICATION FLASHPOINT:

- *"The test environment was delivered 3 weeks late"* → **Project Risk** (threatens project schedule).
- *"The mobile application drains battery rapidly during GPS navigation"* → **Product Risk** (non-functional quality defect).

5.2.3 Product Risk Analysis FL-5.2.3 (K2)

Core Purpose: Provides an awareness of product risks to focus testing in a way that **minimizes residual risk**. It consists of two sub-activities: **Risk Identification** and **Risk Assessment**.

1. Risk Identification:
Generating a comprehensive list of potential risks using brainstorming workshops, stakeholder interviews, expert checklists, and cause-effect diagrams.

2. Risk Assessment:
Categorizing risks, determining likelihood and impact, calculating risk level, and prioritizing risks.

- *Quantitative:* Multiplying numerical likelihood by impact.
- *Qualitative:* Plotting on a **Risk Matrix** (Low, Med, High).

How Product Risk Analysis Influences Testing Scope & Thoroughness:

- **Determines Test Scope:** Higher-risk features receive extensive testing; low-risk areas receive minimal or no testing.
- **Determines Test Levels & Types:** Critical security risks dictate dedicated penetration and vulnerability testing.
- **Determines Test Techniques:** High-risk logic warrants rigorous 3-value BVA and branch coverage; lower risks use exploratory testing.
- **Estimates Effort:** Guides effort allocation toward components with the highest risk ratings.
- **Prioritizes Test Execution:** High-risk tests run first to find critical defects as early as possible.
- **Identifies Non-Testing Actions:** Prompts training, prototyping, or code refactoring to reduce risk upstream.

5.2.4 Product Risk Control FL-5.2.4 (K2)

Core Purpose: Encompasses all measures taken in response to identified and assessed risks. Comprises **Risk Mitigation** (implementing actions to lower risk) and **Risk Monitoring** (tracking effectiveness and discovering emerging risks).

Once risks are assessed, organizations choose from four response options (Veenendaal 2012):

1. Risk Mitigation
Taking active steps (e.g. testing) to reduce likelihood or impact.

2. Risk Acceptance
Consciously deciding to live with the risk without action.

3. Risk Transfer
Passing risk to third parties (insurance, vendor warranty).

4. Contingency Plan
Pre-planned response if the risk occurs (disaster recovery).

Actions Taken to Mitigate Product Risk via Testing:

- Assigning testers with the right experience, domain skills, and technical expertise.
- Applying an appropriate degree of test independence.
- Performing static testing (requirements reviews, code static analysis).
- Selecting rigorous test techniques and setting high coverage thresholds.
- Applying dedicated non-functional test types (load, stress, penetration).
- Conducting extensive automated regression testing.

1.3 Test Monitoring, Test Control and Test Completion

Test Monitoring:
Ongoing activity of gathering information about testing, assessing progress against the test plan, and tracking exit criteria achievement.

Test Control (Control Directives):
Taking corrective actions based on monitoring data to guide testing toward its objectives (e.g. reprioritizing tests when risks become issues, reallocating staff).

5.3.1 Metrics Used in Testing FL-5.3.1 (K1)

Metric Category	Specific Metric Examples & Purpose
Project Progress	Task completion %, resource utilization rate, test effort consumed vs planned.
Test Progress	Test case preparation progress, environment readiness %, test cases run / passed / failed, execution duration.
Product Quality	Availability %, mean time to failure (MTTF), transaction response times, defect density.
Defect Metrics	Number and priority of defects found/fixed, defect density, Defect Detection Percentage (DDP).
Risk Metrics	Residual risk levels, number of high-risk items remaining untested.
Coverage Metrics	Requirements coverage %, code coverage % (statement, branch), user story acceptance criteria coverage.
Cost Metrics	Direct cost of testing, organizational cost of quality (prevention vs failure costs).

5.3.2 Test Reports: Progress vs. Completion Reports FL-5.3.2 (K2)

Dimension	Test Progress Report (Status Report)	Test Completion Report
Frequency & Timing	Generated regularly (daily, weekly, per sprint) during ongoing testing.	Prepared once at project milestones (release, sprint end, test level closure).
Primary Purpose	Supports ongoing test control; informs modifications to schedules and resources.	Summarizes test activity, evaluates product quality, provides closure for stakeholders.
Typical Content	• Testing period & current progress (ahead/behind) • Test metrics (passed/failed, remaining tests) • Blockers/impediments & workarounds • New and changed risks; next period plan	• Executive test summary & product quality evaluation • Deviations from test plan (schedule, budget, effort) • Unmitigated risks and unresolved defects • Documented lessons learned & improvement action items

5.3.3 Communicating Testing Status (FL-5.3.3):

Status communication must be tailored to stakeholder needs: (1) **Verbal communication** (daily stand-ups), (2) **Dashboards** (CI/CD pipelines, Kanban boards, burndown charts), (3) **Electronic channels** (email, chat), (4) **Online documentation** (wikis), and (5) **Formal reports** (ISO/IEC/IEEE 29119-3 templates).

1.4 Configuration Management & Defect Management

5.4 Configuration Management (CM) FL-5.4.1 (K2)

> **Purpose:** Provides a discipline for identifying, controlling, tracking, and versioning all testware items (test plans, test cases, scripts, results, environments) as **configuration items**.

- **Baselines:** An approved configuration item becomes a baseline that can only be altered through formal change control. Enables reverting to previous configurations to reproduce earlier test results.
- **Traceability Support:** CM ensures all test items (parts of test object) and software documentation items are uniquely identified and referenced unambiguously throughout testware.
- **DevOps Integration:** Automated configuration management is an integral foundation of automated CI/CD deployment pipelines.

5.5 Defect Management FL-5.5.1 (K3)

An established defect management process provides a standardized workflow from defect discovery through analysis, classification, remediation, and closure.

DEFECT LIFECYCLE WORKFLOW

1. Log Anomaly → 2. Analyze & Classify → 3. Assign / Fix → 4. Confirmation Test → 5. Close Report

Severity vs. Priority: The Fundamental Distinction:

Severity (Technical Impact):
The degree of impact the defect has on system operation or user requirements (e.g. Minor, Major, Critical, Blocker).

Priority (Business Urgency):
The urgency with which the defect must be fixed based on business goals and release timelines (e.g. Low, Medium, High, Immediate).

Mandatory Fields in a Defect Report (ISO/IEC/IEEE 29119-3 / FL-5.5.1 — K3):

Field	Description & Purpose
Unique Identifier	Unique tracking ID automatically assigned by the defect management tool.
Title & Summary	Short, clear, descriptive summary of the observed anomaly.
Date, Author, Role	Date of discovery, name of person logging defect, and their organizational role.
Test Object & Environment	Exact build version, module ID, hardware platform, OS, and browser details.
Context of Defect	Test activity (e.g. regression), test case executed, checklist used, test data.
Reproduction Steps	Step-by-step instructions to reproduce failure, including logs, dumps, screenshots, or videos.
Expected vs. Actual	Explicit statement of what should have occurred vs what actually occurred.
Severity & Priority	Severity (degree of technical harm) and Priority (urgency to resolve).
Status & References	Status (New, Open, In Progress, Fixed, Awaiting Retest, Closed) and linked Test Case ID.

2. High-Yield Exam Flashpoints & Key Distinctions

TOP 10 ISTQB CTFL v4.0.1 CHAPTER 5 EXAM TRAPS & DISTRACTORS:

1. **"The Test Pyramid states top-layer tests have highest isolation"** → **FALSE!** The top layer (UI/E2E) has the *lowest* isolation and slowest execution. Bottom unit tests have the highest isolation.
2. **"Usability testing belongs in Quadrant Q4"** → **FALSE!** While performance and security belong in Q4 (technology-facing), usability testing belongs in **Q3** (business-facing, critique product).
3. **"Running out of time is never a valid exit criterion"** → **FALSE!** Running out of time or budget IS a valid exit criterion if stakeholders formally review and accept the residual risk.
4. **"Higher priority tests always run before lower priority tests"** → **FALSE!** If a higher-priority test is dependent on a lower-priority test, the lower-priority test must execute first!
5. **"Severity and Priority are identical"** → **FALSE!** Severity is technical impact (system crash); Priority is business urgency (marketing deadline). A spelling error on the homepage has low severity but high priority!
6. **"Three-point estimation uses simple unweighted average"** → **FALSE!** It uses a weighted arithmetic mean: **E = (a + 4m + b) / 6**.
7. **"All anomalies reported in testing are defects"** → **FALSE!** Anomalies can turn out to be false-positives, duplicate reports, or change requests.
8. **"Project risks and product risks are managed identically"** → **FALSE!** Project risks threaten schedule/budget; product risks threaten user quality characteristics.
9. **"Test Progress Reports occur only once at the end of testing"** → **FALSE!** Progress reports occur regularly during testing; Test Completion Reports occur once at milestones.
10. **"Test strategy is an examinable section in Chapter 5"** → **FALSE!** The detailed test strategy section was removed from Foundation Level v4.0 (retained only as a keyword).

Rapid Comparison Matrix (Core Management Distinctions)

Concept A	Concept B	Crucial Distinguishing Test
Entry Criteria (DoR)	**Exit Criteria (DoD)**	Entry criteria define *when to start* an activity; Exit criteria define *when an activity is complete*.
Project Risk	**Product Risk**	Project risk impacts *schedule, budget, team*; Product risk impacts *functional correctness & quality attributes*.
Severity	**Priority**	Severity measures *technical impact*; Priority measures *business urgency to fix*.
Risk Mitigation	**Risk Monitoring**	Mitigation implements *actions to reduce risk*; Monitoring *checks effectiveness & tracks emerging risks*.
Q1 (Tech-Support)	**Q4 (Tech-Critique)**	Q1 supports dev with *unit/integration tests*; Q4 critiques product with *performance/security tools*.

High-Yield Memory Mnemonics for Revision:

- **PERT Formulas: E = (a + 4m + b) / 6 | SD = (b – a) / 6**.
- **Pyramid Rule:** *Bottom = Fast, Isolated, Many | Top = Slow, Low Isolation, Few*.
- **Quadrants:** *Q1 = Unit/CI | Q2 = User Story/API | Q3 = Exploratory/Usability | Q4 = Non-Functional/Smoke*.
- **Risk Responses:** *Mitigate (Test) • Accept • Transfer • Contingency*.

3. Complete Chapter 5 Keyword Lexicon (All 26 Official Terms)

Defect Management The process of recognizing, recording, classifying, investigating, resolving and disposing of defects.

Defect Report Documentation of the occurrence, nature, and status of a defect or anomaly detected during testing.

Entry Criteria The set of conditions for officially starting a defined task or test activity (Definition of Ready).

Exit Criteria The set of conditions for officially completing a defined task or test activity (Definition of Done).

Product Risk A risk impacting the quality of a product (quality characteristics, incorrect behavior, non-compliance).

Project Risk A risk related to the management and control of the project (schedule, budget, staffing, suppliers).

Risk A factor that could result in future negative consequences, characterized by likelihood and impact.

Risk Analysis The overall process comprising risk identification and risk assessment.

Risk Assessment The process of categorizing identified risks, evaluating likelihood and impact, and determining risk level.

Risk Control The overall process comprising risk mitigation and risk monitoring.

Risk Identification The process of recognizing, identifying, and describing potential risks.

Risk Level The qualitative or quantitative measure of a risk, determined by risk likelihood and risk impact.

Risk Management The application of coordinated activities to direct and control an organization with regard to risk.

Risk Mitigation The process through which decisions are reached and protective measures are implemented to reduce risk.

Risk Monitoring The process of tracking identified risks, verifying mitigation effectiveness, and finding emerging risks.

Risk-Based Testing Testing in which test activities are selected, prioritized, and managed based on risk analysis and control.

Test Approach The implementation of test strategy for a specific project, defining levels, types, and techniques.

Test Completion Report A report produced at completion milestones summarizing test activities and product quality evaluation.

Test Control Developing and applying corrective actions to guide a test project on track when monitoring reveals deviations.

Test Monitoring The ongoing checking of test activities and comparison of actual progress against the test plan.

Test Plan Documentation describing the test objectives, resources, approach, schedule, and constraints of a test project.

Test Planning The activity of establishing or updating the scope, approach, resources, and schedule of testing activities.

Test Progress Report A report published at regular intervals during testing to communicate current progress and impediments.

Test Pyramid A graphical model illustrating that different tests have different granularity, isolation, and execution speeds.

Test Strategy Documentation of the organization's high-level alignment and rules for testing across projects.

Testing Quadrants A model grouping test levels with appropriate test types across business/technology and support/critique dimensions.

STUDENT NOTES & EXAM FLASHCARD REMINDERS (Handwritten Study Space)

4. Realistic ISTQB Practice Exam Questions & Detailed Rationales (Part 1)

Question 1 [K3 - FL-5.1.4]: Using the Three-Point Estimation technique, experts provide the following test execution estimates: Most Optimistic (a) = 10 hours, Most Likely (m) = 16 hours, and Most Pessimistic (b) = 34 hours. What are the final weighted Estimate (E) and Standard Deviation (SD)?

A) E = 18 hours, SD = 4 hours

B) E = 20 hours, SD = 4 hours

C) E = 18 hours, SD = 6 hours

D) E = 16 hours, SD = 2 hours

Correct Answer: A

Rationale: Using PERT formulas:

• **E** = (a + 4m + b) / 6 = (10 + 4(16) + 34) / 6 = (10 + 64 + 34) / 6 = 108 / 6 = **18 hours**

• **SD** = (b − a) / 6 = (34 − 10) / 6 = 24 / 6 = **4 hours**

Thus, the final estimate is **18 ± 4 hours** (Section 5.1.4).

Question 2 [K3 - FL-5.1.5]: Test suite execution prioritization is being planned. Test Case TC1 has High risk priority, but requires prerequisite database records generated by Test Case TC2, which has Low risk priority. Test Case TC3 has Medium risk priority and is completely independent. In what order should the tests execute?

A) TC1 → TC3 → TC2

B) TC2 → TC1 → TC3

C) TC3 → TC1 → TC2

D) TC1 → TC2 → TC3

Correct Answer: B

Rationale: Section 5.1.5 states that while priority ordering is desired, test dependencies take precedence: "If a test case with a higher priority is dependent on a test case with a lower priority, the lower priority test case must be executed first." Because TC1 depends on TC2, TC2 must run before TC1. TC3 (Medium) runs after the High-priority workflow completes.

Question 3 [K1 - FL-5.1.6]: According to the Test Pyramid model, which of the following statements correctly characterizes tests located at the TOP layer?

A) They have the highest test isolation and fastest execution time.

B) They check small pieces of isolated functionality, requiring many tests for coverage.

C) They are complex, end-to-end tests that generally have lower test isolation and slower execution time.

D) They represent unit tests integrated into continuous integration builds.

Correct Answer: C
Rationale: Section 5.1.6 establishes that in the test pyramid, the higher the layer, the lower the test granularity, the lower the test isolation (higher dependency on other elements), and the higher the execution time.

Question 4 [K2 - FL-5.1.7]: In the Testing Quadrants model, which quadrant contains exploratory testing and usability testing?

A) Q1 (Technology facing, support the team)

B) Q2 (Business facing, support the team)

C) Q3 (Business facing, critique the product)

D) Q4 (Technology facing, critique the product)

Correct Answer: C
Rationale: Quadrant Q3 contains business-facing tests that critique the product, specifically exploratory testing, usability testing, and user acceptance testing (Section 5.1.7).

4. Realistic ISTQB Practice Exam Questions & Detailed Rationales (Part 2)

Question 5 [K2 - FL-5.2.2]: Which of the following is an example of a PROJECT RISK rather than a product risk?

A) The web application takes more than 10 seconds to process checkout transactions under peak load.

B) A third-party test tool vendor goes bankrupt, causing delays in test environment readiness.

C) The payment calculation module incorrectly rounds tax on international currency exchanges.

D) The software interface fails to comply with industry accessibility regulations.

Correct Answer: B
Rationale: Third-party supplier failure causing delivery delays is a project risk affecting schedule and resources (Section 5.2.2). Options A, C, and D are product quality characteristics (performance efficiency, functional correctness, regulatory compliance).

Question 6 [K2 - FL-5.3.2]: What is the primary difference between a Test Progress Report and a Test Completion Report?

A) Test progress reports evaluate exit criteria against business goals, while test completion reports issue control directives.

B) Test progress reports are generated regularly during testing to support test control, while test completion reports are produced at project milestones to evaluate overall quality.

C) Test progress reports contain documented lessons learned, while test completion reports list next week's test schedule.

D) Test progress reports are authored exclusively by project managers, while completion reports are written by external auditors.

Correct Answer: B
Rationale: Section 5.3.2 defines that test progress reports support ongoing test control and adjustments during testing, while test completion reports consolidate experience, evaluate quality against original exit criteria, and document lessons learned at project milestones.

Question 7 [K2 - FL-5.4.1]: How does Configuration Management directly support the testing process?

A) By executing automated regression test suites whenever developers check in source code.

B) By ensuring all test items and testware are uniquely identified, version-controlled, and tracked to maintain traceability.

C) By automatically calculating PERT three-point estimates for sprint user stories.

D) By determining whether risks should be transferred to third-party vendors.

Correct Answer: B
Rationale: Section 5.4 states that Configuration Management ensures all configuration items (including test items) are uniquely identified, version controlled, tracked for changes, and related to other items so traceability is maintained throughout the test process.

Question 8 [K3 - FL-5.5.1]: A tester discovers that clicking "Print Invoice" causes an unhandled application crash on iOS devices, though Android devices function normally. Which information is CRITICAL to include in the defect report to ensure developers can reproduce and fix the bug?

A) The tester's salary, the test manager's approval signature, and the total cost of quality.

B) A vague note stating "printing doesn't work on mobile".

C) Steps to reproduce, expected vs actual results, device OS version, and relevant system crash logs.

D) The Wideband Delphi estimate for rewriting the iOS application.

Correct Answer: C
Rationale: Section 5.5 outlines that a defect report logged during dynamic testing must include clear reproduction steps, test environment details (OS version), expected vs actual results, and supporting logs/screenshots.

Part 2: Additional Questions for Remaining Learning Objectives

Question 9 [K2 - FL-5.1.1]: Which of the following is typical content of a test plan?

A) The source code of the unit tests and the detailed list of defects found.

B) A risk register, the test approach (levels, types, techniques, entry and exit criteria), stakeholders, budget and schedule.

C) Salary reviews and annual appraisals of the testers.

D) Only the final test results and the release decision.

Correct Answer: B
Rationale: A test plan describes the test objectives, resources and processes. Typical content (ISO/IEC/IEEE 29119-3): context (scope, objectives, test basis), assumptions and constraints, stakeholders, communication, risk register (product and project risks), test approach (levels, types, techniques, deliverables, entry/exit criteria, metrics, test data and environment needs), and budget and schedule (Section 5.1.1).

Question 10 [K1 - FL-5.1.2]: Which of the following is a way in which a tester adds value to iteration and release planning?

A) Writing the production code for the user stories.

B) Approving the release on behalf of the product owner.

C) Deciding the project budget and the sprint length alone.

D) Taking part in user story refinement (testable stories with acceptance criteria), performing product risk analysis, estimating test effort and planning the test approach.

Correct Answer: D
Rationale: In release and iteration planning a tester helps to write testable user stories and acceptance criteria, performs product quality risk analysis, estimates the test effort, identifies functional and non-functional aspects to be tested and plans the testing within the iteration (Section 5.1.2).

Question 11 [K2 - FL-5.1.3]: Statement 1: "The test environment and test data are available." Statement 2: "The planned coverage level has been reached and the number of open critical defects is within the agreed limit." How are they correctly classified?

A) Statement 1 is an entry criterion; statement 2 is an exit criterion (in agile: Definition of Ready and Definition of Done).

B) Statement 1 is an exit criterion; statement 2 is an entry criterion.

C) Both statements are entry criteria.

D) Both statements are exit criteria.

Correct Answer: A
Rationale: Entry criteria are the preconditions for starting an activity (testable requirements, resources such as environments and data, initial quality of the test object). Exit criteria define what must be achieved to declare an activity complete (planned tests executed, coverage level reached, unresolved defects within an agreed limit). In agile projects they correspond to the Definition of Ready and the Definition of Done (Section 5.1.3).

Question 12 [K1 - FL-5.2.1]: How is the risk level of a risk determined?

A) Only by the cost of the mitigation actions.

B) Only by the number of testers available.

C) By combining the likelihood of the risk occurring and its impact (for example with a risk matrix or numeric ratings).

D) Only by the duration of the project.

Correct Answer: C
Rationale: The risk level is determined by risk likelihood (the probability that the risk occurs, greater than zero and less than one) and risk impact (the consequences of its occurrence). It can be expressed qualitatively (high/medium/low) or quantitatively (Section 5.2.1).

Question 13 [K2 - FL-5.2.3]: A product risk analysis rates the payment module as high risk and the "About us" page as low risk. How should these results influence testing?

A) Both should receive identical effort to stay fair.

B) The payment module gets more thorough techniques, higher coverage and earlier testing; the "About us" page gets minimal effort.

C) The payment module should be skipped because high-risk items are too expensive to test.

D) The "About us" page should be tested first because it is the easiest.

Correct Answer: B
Rationale: Product risk analysis results are used to decide the test scope, the test levels and types, the test techniques and coverage, the test effort per task, and the priorities so that critical defects are found as early as possible; they may also point to non-testing activities that reduce risk. Higher risk means more thorough and earlier testing (Section 5.2.3).

Question 14 [K2 - FL-5.2.4]: A product risk analysis identifies a high security risk. Which of the following is an appropriate response measure?

A) Reduce the test effort to meet the schedule, because security defects are rare.

B) Remove the security requirement from the test basis.

C) Assign testers with suitable security skills, apply security testing and static analysis, and increase the independence of testing.

D) Wait for a production incident and fix the problem afterwards.

Correct Answer: C
Rationale: Measures in response to analyzed product risks include: choosing testers with the right experience and skills for the risk type, applying an appropriate level of independence, performing reviews and static analysis, applying suitable test techniques and coverage levels, applying test types addressing the affected quality characteristics (such as security testing), and performing dynamic testing including regression testing (Section 5.2.4).

Question 15 [K1 - FL-5.3.1]: Which of the following is a TEST PROGRESS metric?

A) Mean time to failure of the released product.

B) The number of test cases run and not run, passed and failed.

C) Defect density.

D) Cost of quality.

Correct Answer: B
Rationale: Test progress metrics include the number of test cases run / not run, passed / failed, test case implementation progress, test environment preparation progress and test execution time. Mean time to failure is a product quality metric, defect density is a defect metric and cost of quality is a cost metric (Section 5.3.1).

Question 16 [K2 - FL-5.3.3]: A distributed agile team (three countries) uses a CI/CD pipeline. Which is the most suitable way to communicate the status of testing continuously?

A) Printing a monthly test report and sending it by post.

B) Hallway conversations only.

C) Waiting until the end of the project to present the test completion report.

D) Dashboards (for example CI/CD dashboards, task boards and burn-down charts), supported by chat, online documentation and short verbal updates.

Correct Answer: D
Rationale: Ways to communicate test status include verbal communication, dashboards (CI/CD dashboards, task boards, burn-down charts), electronic channels (e-mail, chat), online documentation and formal test reports. The choice depends on the audience, project risk, organizational standards and whether the team is distributed. A distributed agile team benefits from continuous, shared channels such as dashboards and chat (Section 5.3.3).